

## FORMOSA, Argentina

WMO: 871620

Lat: 26.2125S

Lon: 58.2292W

Elev: 197

StdP: 14.59

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 39.8       | 43.1 | 33.2                          | 28.0 | 50.7 | 35.9 | 31.1 | 50.1 | 28.9                  | 72.7 | 24.7 | 74.0 | 4.2                   | 230  | 0.399 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 19.4                   | 100.0           | 76.6 | 97.6 | 76.8 | 95.4 | 76.7 | 81.5                | 91.6 | 80.6 | 90.7 | 79.7 | 89.7 | 13.5                 | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.0                            | 151.7 | 86.5 | 78.1 | 146.9 | 85.8 | 77.1 | 142.1 | 85.0 | 45.6          | 91.9 | 44.5 | 90.9 | 43.5 | 90.2 | 91.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 23.5          | 20.5 | 18.0 | 33.9                       | 104.1 | 2.9                | 2.0 | 31.8                                               | 105.5 | 30.2       | 106.6 | 28.6       | 107.7 | 26.5       | 109.2 |    |    |
| (5)               |      |      | 33.2                       | 84.3  | 2.8                | 2.3 | 31.1                                               | 86.0  | 29.4       | 87.3  | 27.9       | 88.5  | 25.8       | 90.2  |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 73.1  | 82.7  | 81.5  | 78.7  | 73.8  | 66.9  | 63.9  | 62.9  | 66.1  | 69.3  | 74.6  | 76.9  | 80.7  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.94  | 4.24  | 4.81  | 5.24  | 6.89  | 7.59  | 8.82  | 9.69  | 9.86  | 8.88  | 6.91  | 5.66  | 4.96  | (7)  |
| (8)  |                                                                                  | HDD50       | 14    | 0     | 0     | 0     | 0     | 0     | 3     | 8     | 3     | 0     | 0     | 0     | 0     | (8)  |
| (9)  |                                                                                  | HDD65       | 552   | 0     | 0     | 0     | 12    | 70    | 131   | 166   | 111   | 54    | 6     | 2     | 0     | (9)  |
| (10) |                                                                                  | CDD50       | 8453  | 1012  | 881   | 891   | 713   | 524   | 420   | 408   | 503   | 579   | 764   | 807   | 951   | (10) |
| (11) |                                                                                  | CDD65       | 3517  | 547   | 461   | 427   | 275   | 128   | 98    | 101   | 147   | 184   | 305   | 358   | 486   | (11) |
| (12) |                                                                                  | CDH74       | 35848 | 6443  | 5045  | 4202  | 2311  | 848   | 575   | 752   | 1505  | 1922  | 2949  | 3701  | 5595  | (12) |
| (13) |                                                                                  | CDH80       | 16707 | 3312  | 2462  | 1931  | 937   | 256   | 141   | 238   | 656   | 885   | 1357  | 1708  | 2824  | (13) |
| (14) | Wind                                                                             | WSAvg       | 8.1   | 7.8   | 7.4   | 7.0   | 7.2   | 7.1   | 7.7   | 8.2   | 9.1   | 9.4   | 9.5   | 8.7   | 8.2   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 55.6  | 6.6   | 5.5   | 5.6   | 6.2   | 4.2   | 2.6   | 1.7   | 1.9   | 3.2   | 5.4   | 6.4   | 6.5   | (15) |
| (16) |                                                                                  | PrecMax     | 80.3  | 13.3  | 25.7  | 14.1  | 15.6  | 10.7  | 7.4   | 6.5   | 7.7   | 8.8   | 11.5  | 18.4  | 16.7  | (16) |
| (17) |                                                                                  | PrecMin     | 32.5  | 0.3   | 0.2   | 0.3   | 0.7   | 0.5   | 0.2   | 0.0   | 0.0   | 0.0   | 0.6   | 0.9   | 1.4   | (17) |
| (18) |                                                                                  | PrecStd     | 10.5  | 3.3   | 3.9   | 3.0   | 3.9   | 2.6   | 1.7   | 1.5   | 1.8   | 2.1   | 2.8   | 3.5   | 3.5   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 100.8 | 101.0 | 99.2  | 95.3  | 90.2  | 86.9  | 89.3  | 95.4  | 99.3  | 101.4 | 100.5 | 101.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 77.5  | 77.5  | 76.5  | 77.1  | 75.0  | 72.8  | 70.5  | 70.7  | 74.0  | 77.2  | 76.5  | 75.6  | (20) |
| (21) |                                                                                  | 2%          | DB    | 98.5  | 98.2  | 96.4  | 92.6  | 86.3  | 84.2  | 86.0  | 91.6  | 94.9  | 96.9  | 96.3  | 98.2  | (21) |
| (22) |                                                                                  |             | MCWB  | 77.1  | 77.8  | 77.3  | 76.2  | 72.9  | 71.4  | 70.0  | 70.9  | 72.8  | 76.4  | 76.4  | 76.6  | (22) |
| (23) |                                                                                  | 5%          | DB    | 96.4  | 95.1  | 93.2  | 89.4  | 82.5  | 80.8  | 82.5  | 87.8  | 89.7  | 91.8  | 92.9  | 95.2  | (23) |
| (24) |                                                                                  |             | MCWB  | 77.4  | 78.5  | 76.8  | 75.5  | 71.5  | 70.6  | 69.3  | 69.6  | 71.6  | 75.2  | 75.2  | 76.7  | (24) |
| (25) |                                                                                  | 10%         | DB    | 93.5  | 91.8  | 89.8  | 85.9  | 78.7  | 76.9  | 78.0  | 82.7  | 84.7  | 87.5  | 89.3  | 92.9  | (25) |
| (26) |                                                                                  |             | MCWB  | 77.0  | 77.1  | 76.0  | 74.6  | 70.4  | 69.7  | 67.9  | 68.5  | 69.9  | 73.7  | 73.9  | 76.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 82.5  | 83.1  | 81.6  | 80.6  | 77.5  | 74.7  | 74.3  | 75.3  | 77.8  | 80.5  | 81.6  | 82.1  | (27) |
| (28) |                                                                                  |             | MCDWB | 93.2  | 92.8  | 91.2  | 90.0  | 85.3  | 82.6  | 83.8  | 87.9  | 91.5  | 92.8  | 93.0  | 91.8  | (28) |
| (29) |                                                                                  | 2%          | WB    | 81.1  | 81.0  | 80.0  | 78.9  | 75.4  | 73.1  | 72.1  | 72.9  | 75.5  | 78.6  | 79.5  | 80.8  | (29) |
| (30) |                                                                                  |             | MCDWB | 91.4  | 90.7  | 89.6  | 88.3  | 83.2  | 80.7  | 81.1  | 87.2  | 89.0  | 91.7  | 90.3  | 91.1  | (30) |
| (31) |                                                                                  | 5%          | WB    | 80.1  | 80.0  | 79.0  | 77.5  | 73.5  | 71.7  | 70.2  | 70.8  | 73.4  | 76.8  | 77.9  | 79.7  | (31) |
| (32) |                                                                                  |             | MCDWB | 90.1  | 89.6  | 88.5  | 85.7  | 79.8  | 78.6  | 79.5  | 83.5  | 86.2  | 88.7  | 88.0  | 89.8  | (32) |
| (33) |                                                                                  | 10%         | WB    | 79.2  | 79.0  | 77.7  | 76.0  | 71.7  | 69.9  | 68.4  | 68.8  | 71.3  | 75.0  | 76.3  | 78.6  | (33) |
| (34) |                                                                                  |             | MCDWB | 88.8  | 88.4  | 86.5  | 83.2  | 76.9  | 76.3  | 76.8  | 80.7  | 82.1  | 85.7  | 85.4  | 88.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 19.4  | 18.5  | 18.7  | 17.9  | 16.8  | 16.5  | 19.3  | 21.7  | 21.8  | 19.3  | 19.8  | 19.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 23.3  | 22.4  | 23.2  | 22.0  | 20.4  | 19.0  | 22.3  | 26.2  | 27.9  | 25.4  | 25.6  | 23.9  | (36) |
| (37) |                                                                                  |             | MCWBR | 6.9   | 7.3   | 7.9   | 8.5   | 9.3   | 9.2   | 10.0  | 10.7  | 11.4  | 10.1  | 9.5   | 7.5   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 18.5  | 18.6  | 19.0  | 18.5  | 16.7  | 16.9  | 20.2  | 23.9  | 24.2  | 21.9  | 19.7  | 18.9  | (38) |
| (39) |                                                                                  |             | MCWBR | 7.2   | 7.7   | 7.9   | 8.5   | 8.7   | 9.3   | 10.6  | 11.4  | 11.5  | 10.0  | 9.6   | 7.9   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.409 | 0.409 | 0.401 | 0.401 | 0.380 | 0.401 | 0.390 | 0.497 | 0.565 | 0.492 | 0.412 | 0.414 | (40)  |      |
| (41) |                                                                                  | taud        | 2.405 | 2.411 | 2.416 | 2.394 | 2.415 | 2.342 | 2.362 | 2.011 | 1.845 | 2.113 | 2.354 | 2.374 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 297   | 292   | 285   | 269   | 260   | 245   | 254   | 235   | 233   | 265   | 294   | 296   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 40    | 39    | 37    | 35    | 31    | 32    | 32    | 50    | 64    | 52    | 42    | 42    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2135  | 1956  | 1683  | 1334  | 1030  | 849   | 1005  | 1207  | 1445  | 1712  | 2024  | 2129  | (44)  |      |
| (45) |                                                                                  | RadStd      | 153   | 128   | 120   | 145   | 132   | 84    | 93    | 93    | 164   | 194   | 121   | 144   | (45)  |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | +0.95   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page